CUMULATIVE EXAMINATION
APRIL 20, 2001
PHYSICAL CHEMISTRY

Describe the basic experimental setup of photoelectron spectroscopy, and the information which can be obtained from it. The photoelectron spectrum of cyclopropane, using 58.435 nm He I irradiation, is shown below. The centers of three band, corresponding to ionization from the e',e" and a_1 orbitals are also shown.

(a) What is the lowest ionization potential of cyclopropane (in eV)?

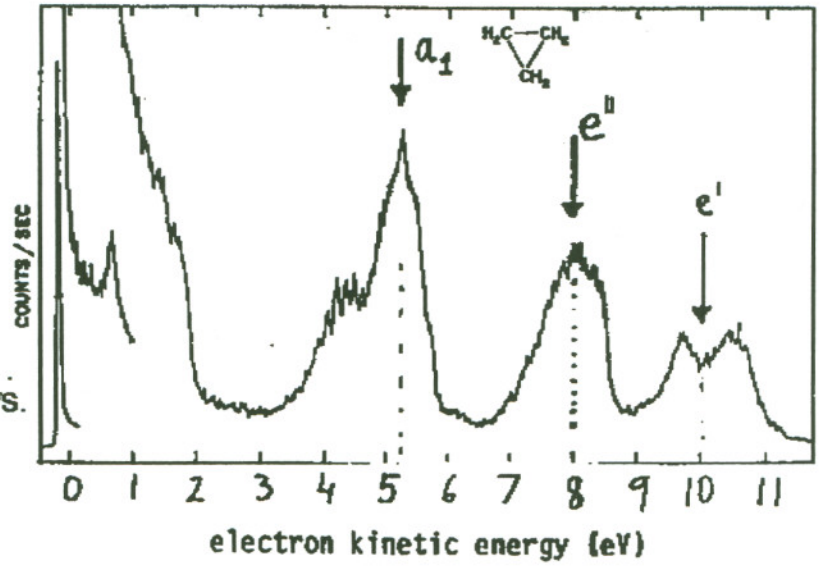
(b) What are the proper term symbols for the three positive ion states (including multiplicity and spatial symmetry)? Which of these states will spontaneously distort from the D_{3h} symmetry of the ground state due to the Jahn-Teller effect?

(c) The ground state of the molecular cation is obtained from removing an electron from the highest occupied molecular orbital. The states formed by removing an electron from a lower-lying orbital are thus excited states of the molecular cation, and will revert to the ground state, for instance by emitting a photon. What are the wavelengths of the photons emitted in this process?

Constants: $c=2.9978*10^{8}$ m/s (speed of light), $h=6.6262*10^{-34}$ Js (Planck's constant), $1eV=1.6020*10^{-19}$ J

Cyclopropane

He I

Photoelectron spectrum of cyclopropane.

J. Basch et al . J. Chem. Phys. 1969